Phase 8 — Functional MVP Development

Goal

Transform the validated prototype into a functional Minimum Viable Product (MVP) by replacing simulated behaviors with working functionality. The objective is to develop an application that users can interact with in a realistic way, where core features operate correctly and the complete user flow is functional.

Step A — Evaluate the Development Environment

Objective

Assess whether the current prototype developed in **Figma Make** can support all required functionalities. If necessary, compare alternative development platforms and justify the final choice.

Questions to Answer

- Which features are already functional?
- Which features are still simulated?
- Can Figma Make support the remaining functionality?
- If not, which platform is the most appropriate for implementation?

Possible platforms include:

- Figma Make
- FlutterFlow
- Bolt.new
- Lovable
- MIT App Inventor

Deliverables

- Development Platform Evaluation
- Justification of the Selected Platform

Step B — Implement Core Functionality

Objective

Replace simulated interactions with real application behavior.

Core functionalities should include:

- User registration or profile creation (if applicable)
- Daily check-in
- Collection of input variables
- Collection of context variables
- Rule-based cognitive state detection
- Personalized recommendation generation
- History tracking
- Insights dashboard

At the end of this step, the application should be capable of completing the full workflow without simulated screens.

Deliverables

- Functional MVP

Step C — Implement Data Storage

Objective

Ensure that user information is stored and retrieved correctly.

Possible stored data include:

- User profile
- Daily check-ins
- Cognitive states
- Recommendation history
- User feedback
- Progress over time

If a backend is used (e.g., Supabase), document how data are organized.

Deliverables

- Functional Database
- Data Structure Documentation

Step D — Implement the Recommendation Engine

Objective

Replace simulated recommendations with a real rule-based decision system.

The application should:

- Receive user inputs
- Interpret collected variables
- Detect the appropriate cognitive state
- Generate the corresponding recommendation
- Save the result to the user's history

The decision process should follow the system architecture defined in previous phases.

Deliverables

- Functional Rule-Based Recommendation Engine
- Decision Logic Documentation

Step E — Integrate and Validate the Complete Workflow

Objective

Verify that every component of the application works together correctly.

Complete workflow:

1. User opens the application.
2. User completes the daily check-in.
3. Data are processed.
4. Cognitive state is identified.
5. Personalized recommendation is generated.
6. Results are saved.
7. Dashboard is updated.

Identify and correct any implementation issues before user testing.

Deliverables

- Functional MVP
- Integration Checklist
- Implementation Notes

Final Deliverables of Phase 8

Required

- Functional MVP
- Development Documentation
- Rule-Based Recommendation System
- Functional Data Storage
- Integration Checklist

Optional

- Source Code
- Database Diagram
- System Architecture Diagram
- Backend Configuration Documentation

Success Criteria

By the end of this phase, the student should be able to answer:

- Does the application perform all core functions without simulation?
- Can users complete the entire workflow successfully?
- Are cognitive states generated automatically based on user inputs?
- Are personalized recommendations generated consistently?
- Is user data stored and retrieved correctly?
- Is the application ready for final user testing?